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# Mac é...ç½® GD32 ä¼€â•ŽžŒƒæ™ç‹‹ ## 1. æ•™ç‹‹ç›®æ‡æœ–æ–†æ¡fèˆ
‘æŽšä½‚ä½•âœ macOs ä,Šé...ç½® GD32
â•Ž•ç‰‰æœă¼€â•ŽžŒƒæ™ç‹‹æƒfi¼€âŽžžŽ“ăYă,«èf¼ăŠİ¼š - ä,â½¿ç” Keilă€, - ä,â¾â•Žèµ–
Windows âŽŸăŽ, IDEă€, - ä½¿ç” GCC â.Yă...é“¼ç¼–è‘ GD32 â°ă»¶ă€, - ä½¿ç” CMake/
Ninja ç®¡ç†ă¥ç‹‹ăž,,ă»ăœ, - ä½¿ç” ST-Link çƒšă½•ă•ĈéâĈē◊ăă»¶ă€, -
ä½,âĈâĈăŽŸăŽ, SDK ç›®ă½•ă•â•Ž »İ¼Ĉă,ç›’æŽŸă½;®æ”¹ăŽŸăšçëµ,,æ–™ăœ
, æœ–é¡ç›®ç›®æ‡èŠç‰‰, ° `GD32F303RCT6`İ¼Ĉă±žăŽ GD32F30x è‹ă–†ăº Cortex-M4F
ç³»ă—ăœ, ## 2. çŽŒƒçƒç»,æ•Ž macOS ä,ŠæŽŒè◊â½¿ç”ăYă,ă.Yă...İ¼š | â.Yă... | ä½œç”
|| --- || Homebrew | macOS âĈ...ç®¡ç†ă™İ¼Ĉç”ăŽă®œœf...ă¼€â•Žă.Yă... || arm-
none-eabi-gcc | Arm Cortex-M èf,æœ°ăºă◊‰ç¼–è‘ă™ || CMake | â¥ç‹‹é...
ç½®ă•ĈăŽæž,,ă»ç”Yăæ◊ă.Yă... || Ninja | è‹æ•æž,,ă»ăŽç‹– || stlink | ST-Link
æŽçæµ.ăœçƒšă½•ăœè–ă»ăžă.Yă... || OpenOCD | â◊–éç‰œ°fê•ă.Yă...İ¼ĈăŽç‹–
â◊–ç”ăŽ GDB è°fê• | ## 3. â®œœf... Homebrew â|,æžœ Mac ä,Šě¿æ²jæœ‰
Homebrewİ¼Ĉă...ă®œœf... Homebrewİ¼š ``sh /bin/bash -c "$(curl -fsSL https://
raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)" `` â®œœf...
â®Ĉæ•âŽç¡;®è®İ¼š ``sh brew --version `` â|,æžœă½ă»ă,âăœİ¼Ĉéœœè|çæ'æ•®
Homebrew â®œœf...æçççœŠŠ `/opt/homebrew/bin` âŠă...¥`PATH`ăœ, Apple Silicon Mac
ă,,èšçè–ă¾,İ¼š ``sh eval "$(/opt/homebrew/bin/brew shellenv)" `` Intel Mac
ă,,èšçè–ă¾,İ¼š ``sh eval "$(/usr/local/bin/brew shellenv)" `` ## 4. â®œœf... GD32
ă¼€â•Žă.Yă...é“¼æ‰‰şçİ¼š ``sh brew install cmake ninja arm-none-eabi-gcc stlink open-
ocd `` â®œœf...â®Ĉæ•âŽžéâĈē◊İ¼š ``sh arm-none-eabi-gcc --version cmake --version
ninja --version st-info --version st-flash --version openocd --version ``
â|,æžœè¿™ăöă½ă»œèf½æfă,,è¼“ă†°ç‰‰æœ–ă◊İ¼Ĉē–æŽšăYç¡ă.Yă...
·é“¼ă²ç»â®œœf...â®Ĉæ•ăœ, ## 5. è¿žæŽŸ ST-Link âº† ST-Link è¿žæŽŸă°
Macİ¼Ĉă¶è¿žæŽŸă¼€â•Žæç; SWD æŽŸăœfăœ, â,,èšç SWD æŽŸç;İ¼š | ST-Link |
ă¼€â•Žæç; || --- || SWDIO | SWDIO || SWCLK | SWCLK || GND | GND || 3.3V | 3.3V
â◊,èœfç”µăŽç | NRST | â◊–éç‰œİ¼Ĉē¿žæŽŸăŽăœă½çæ’ă◊–éç | æ³»æ,,çİ¼š - `3.3V`
éœšă,ç”ăŽç”µă½ă◊,èœfi¼Ĉă,â,œăšç”™ă¼€â•Žæç;ă¾ç”µİ¼Ĉă...ă½“çœ‹ ST-Link
ăžăă•ă•Ĉăă¼€â•Žæç;ă¾ç”µæ–ă¼€â•Žăœ, - â½...è¡»ă...±ăœºİ¼Ĉăă³ ST-Link çš,, `GND`
è|ç¿žæŽŸă¼€â•Žæç; `GND`ăœ, - â|,æžœă,ç¿žæŽŸ `NRST`İ¼Ĉē`st-flash`
ă»â◊â◊–èf½éœşç¿ž½ă»¶ăœçă½çă.Yăă½œİ¼Ĉăă½çă–Yăç–ă¼šæçççœ° `NRST is not
connected`ăœ, ## 6. éâĈē◊ ST-Link â•ĈēŠç‰‰è¿žæŽŸ æ‰‰şçİ¼š ``sh st-info --probe ``
æfă,,æf...ă†µă,ă¼šçœçăºç±ă¼ă¼è¼“ă†°İ¼š ``text Found 1 stlink programmers version:
V2J37S7 serial: XXXXXXXXXXXX flash: 262144 (pagesize: 2048) sram: 65536 chipid: 0x414
dev-type: F1xx_HD `` è†çç,'ç¡;®è®İ¼š - `Found 1 stlink programmers`İ¼šST-Link
ă²èç«è–†ăăăœ, - `flash: 262144`İ¼šFlash ä,° 256 KiBăœ, - `sram: 65536`İ¼šST-Link â...
¼ă®ăâççè®®æŠŸăŠăœİ¼İ¼Ĉă,ç›’æŽŸă½œă°,é“¼æŽŸè,,šæœ– SRAM âşşă◊ăœ, - `dev-
type: F1xx_HD`İ¼šGD32F303RCT6 ä,Ž STM32F1 é‹ă–†ăºçƒšă½•âççè®®ă...
¼ă®İ¼Ĉēstlink ä½šè¿™æ –è†ăăăœ, æœ–æç;ăŽŸăŽ, Keil â¥ç‹‹ăœæŽ
`IRAM(0x20000000,0x0C000)`İ¼ĈăâŽé™...é“¼æŽŸè,,šæœ–ă”æĈ‰ 48 KiB SRAM é...
ç½®ăœ,è¥æĈ‰ 64 KiB è®¼ç½® `_estack =
0x20010000`İ¼ĈăâŠš–è¶æµă¼šă»æ`è¡è¶šç•Ĉèš|ăĈ` HardFaultăœ
, â|,æžœæ²jæœ‰è–†ăăă° ST-Linkİ¼ĈăœœœăYŸİ¼š - USB
çº¿æ–ă◊ăœ–æĈæ•œœâ¼ă¼è¼“ăœ, - ST-Link æ–ă◊ă¾ç”µăœ, - macOS æ–ă◊ăœŽăö°
USB è®¿é–®æœœœ™ăœ, - æ–ă◊ăœœ‰ă...¶ă–è½ă»¶ăç”ăº† ST-Linkăœ
, â|,æžœè–†ăăă° ST-Link ä½çè–†ăăă,â°èŠç‰‰İ¼ĈăœœœăYŸİ¼š - SWDIO/SWCLK
æ–ă◊ăœŽŸăœăœ, - GND æ–ă◊è¿žæŽŸăœ, - â¼€â•Žæç;æ–ă◊ă¾ç”µăœ, -
èŠç‰‰æ–ă◊èççăă»ă»ă...é– SWDăœ, - â½...è|çæ–
¶æĈ‰ă½çăœçă½çă†çœ‰şçİ¼ĈăŽçæµ.ăœ, ## 7. â††ăº GD32 SDK
æœ–é¡ç›®ă²æœ‰ăŽŸăŽ,èµ,,æ–™ç›®ă½•İ¼š ``text .../GD32F303Rxt6 KIT `` SDK
ă½¿ç”è–ă¾,İ¼š ``text GD32F303Rxt6 KIT/3. SDK/GD32F303RCT6 Demo/Library/
GD32F30x_Firmware_Library_V2.1.0/Firmware `` è¿™ă,ç›®ă½•ăĈă...â◊ăİ¼š - CMSIS
æ.â|făœ–ăă»ăœ, - GD32F30x è®¼ăºă†æ–ăžăœ, - `system gd32f30x.c`
```

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ç³»ç»Yâ^âšâCE-æ-†â»¶â€, - GD32 æ †â††â-è®¼â°“æ°ç ðâ€, â°è®¼âŽYâ™i¼š -
âŽYâŽ, SDK â°âè»â¼•ç””â€, - ä, ðâœâŽYâŽ, KIT ä, æ””â»£ç ðâ€, - è†ââ±çš,,â·¥ç<æ-
†â»¶æ”³¼âœç<ç<<ç>®â¼½i¼CEä¼¼â¼,æœ-é¼ç>çš,, `firmware/â€
, è¼™æ·â°ðšâð-ä»¥ä¼ð-èâŽYâš<è¼,,æ-™âð-â¼žæ°i¼CEâðŽç»âð†ç°§ SDK æ-
¶âiYâ®æ”æ”è³¼fâ·®â¼,â€, ## 8. â°ç<<æŽ”èððâ·¥ç<ç>“æž,, æŽ”èððâ·¥ç<ç>“æž,i¼š
`text firmware/â”œâ”Êâ”Ê CMakeLists.txt â”œâ”Êâ”Ê cmake/â”, â””â”Êâ”Ê arm-none-eabi-
gcc.cmake â”œâ”Êâ”Ê include/â”, â”œâ”Êâ”Ê board.h â”, â”œâ”Êâ”Ê gd32f30x_libopt.h
â”, â”œâ”Êâ”Ê stdbool.h â”, â”œâ”Êâ”Ê stdint.h â”, â””â”Êâ”Ê systick.h â”œâ”Êâ”Ê ld/
â”, â””â”Êâ”Ê GD32F303RCT6.ld â”œâ”Êâ”Ê src/â”, â”œâ”Êâ”Ê board.c â”, â”œâ”Êâ”Ê
gd32f30x_it.c â”, â”œâ”Êâ”Ê main.c â”, â””â”Êâ”Ê systick.c â””â”Êâ”Ê startup/â””â”Êâ”Ê
startup_gd32f30x_hd_gcc.S `` â°ð,,æ-†â»¶â¼œç””i¼š | æ-†â»¶ | ä¼œç”” | --- | --- |
`CMakeLists.txt | â®šä¼œâ·¥ç<æ°ç ðâ€ðâœðæ-†â»¶è-â¼,â€ç¼-
è-â°ð,æ°â€ðè-“¼æŽYâð,æ°â°CEè³¼“â†æ-†â»¶ | | `cmake/arm-none-eabi-gcc.cmake` |
æCE†â®šä¼œð°ç¼-è-â™”ä,° `arm-none-eabi-gcc` | | `ld/GD32F303RCT6.ld` | â®šä¼œ Flash/
RAM âœâ°â°â€ðâ°ð-âš”â...¥â°£â°CEâ®µâ,fâ±€ | | `startup/startup_gd32f30x_hd_gcc.S` |
GNU â°ð-âš”æ-†â»¶â¼â°CEä,æ-â°ð-è†ðè¼ | | `include/gd32f30x_libopt.h` | €€
%œâ©éœèè¼â°ç””çš,, GD32 â°-è®¼¼â””â°æ-†â»¶ | | `src/board.c` | æ¼ç°§
GPIOâ€USART â°ðâšâCE- | | `src/systick.c` | 1 ms ç³»ç»Yèš,æ<ðâ°CEâ»¶æ—¶ | | `src/main.c`
| â°°ç””â¼»èèè¼¼ | ## 9. CMake â·¥â...é“¼æ-†â»¶ | | `firmware/cmake/arm-none-eabi-gcc.cmake`
ç””â°Žâ°Sè-%œ CMakei¼šè¼™æ””ä,€ä,â€£,æœâ°â°ðâ°%œ¼¼-è-â·¥ç<i¼CEä,ðæ” macOS
æœ-æœ°ç<â°ðâ€, æ ,â¼fè...ðç¼®i¼š ``cmake set(CMAKE_SYSTEM_NAME Generic)
set(CMAKE_SYSTEM_PROCESSOR arm) set(CMAKE_TRY_COMPILE_TARGET_TYPE
STATIC_LIBRARY) find_program(ARM_NONE_EABI_GCC arm-none-eabi-gcc REQUIRED)
find_program(ARM_NONE_EABI_OBJCOPY arm-none-eabi-objcopy REQUIRED)
find_program(ARM_NONE_EABI_SIZE arm-none-eabi-size REQUIRED)
set(CMAKE_C_COMPILER "${ARM_NONE_EABI_GCC}") set(CMAKE_ASM_COMPILER
"${ARM_NONE_EABI_GCC}") set(CMAKE_OBJCOPY "$
{ARM_NONE_EABI_OBJCOPY}") set(CMAKE_SIZE "${ARM_NONE_EABI_SIZE}") ``
â...³é”®ç,i¼š - `Generic` è¼ç°ç>®æ †â,ðæ” macOSâ€Linux è¼™ç±»æ“ðâ¼œç³»ç»Yâ€, -
`STATIC_LIBRARY` è¼ä...ð CMake âœé...
ðç¼®è”¶æ®µâ°ðè-é“¼æŽYâð-æ°%šè¼CEç<â°ðâ€, - `arm-none-eabi-gcc` â°CEæ—¶ç””â°Ž C
â°CEâ±†ç¼-æ-†â»¶â¼â€, ## 10. èšç°%ð†ç¼¼-è-â°ð,æ° GD32F303RCT6 æ” Cortex-
M4Fi¼CEâ° æ°ç¼¼-è-â°ð,æ°ä,°i¼š ``text -mcpu=cortex-m4 -mthumb -mfpv4-sp-d16
-mfloat-abi=hard `` â·¥ç<â®ðâ®šä¼œi¼š ``text GD32F30X_HD
USE_STDPERIPH_DRIVER HXTAL_VALUE=8000000 `` è”æ”Ži¼š -
`GD32F30X_HD`i¼šè€œ%œâ© GD32F30x è<â”†â°âž<â°â€, -
`USE_STDPERIPH_DRIVER`i¼šâ°ç””æ †â††â-è®¼â°“â€, -
`HXTAL_VALUE=8000000`i¼šâ°-èf”æ™¶æCEæ° 8 MHz é...ðç¼®â€, ## 11.
é“¼æŽYè,,šæœ-é...ðç¼® GD32F303RCT6 Flash/SRAM â,fâ±€i¼š ``ld FLASH(rx) :
ORIGIN = 0x08000000, LENGTH = 256K RAM(xrw) : ORIGIN = 0x20000000, LENGTH =
48K `` â°ð-âš”æ-è¼¶i¼š ``ld_estack = ORIGIN(RAM) + LENGTH(RAM); ``
é“¼æŽYè,,šæœ-éœèè¼â°ç,çð†i¼š - `isr_vector`i¼šä,æ-â°ð-è†ðè¼i¼CEä¼...è¼»æ”³¼âœ Flash
è¼·âš<âœâ°â°â€, - `text`i¼šç<â°ðâ»£ç ði¼CEæ”³¼â...¥ Flashâ€, -
`.rodata`i¼šâ°ð-»â,è†ði¼CEæ”³¼â...¥ Flashâ€, - `.data`i¼šâ°â°ðâš<âCE-â...â±
Êâ°è†ði¼CEè¼ðè¼CEâœ RAMi¼CEâ°ðâ¼¼âæ”³¼âœ Flashâ€, - `bss`i¼šæœâ°â°ðâš<âCE-â...
”â±Êâ°è†ði¼CEè¼ðè¼CEâ°%ðæ,...é¼â€, - heap/stack ä¼ç°™â°â°Yâ€, ## 12. â°ð-âš”æ-
†â»¶è...ðç¼® â°ð-âš”æ-†â»¶è”èYèèšç°%ðâ,šç”µâðŽçš,,æœææ—œæYâ°ðâš<âCE-â€
, â°ð-âš”æµðç<i¼š 1. ä»Žä,æ-â°ð-è†ðè¼è-»âð-â°ðâš<æ-è¼¶â€, 2. è³è¼-â°
`Reset_Handler`â€, 3. æ<èè`data` æ®µâ° RAMâ€, 4. æ,...é¼`bss` æ®µâ€, 5. è°fç””
`SystemInit()` é...ðç¼®ç³»ç»Yæ—è”èYâ€, 6. è°fç”” `main()` â€, â¼,æžœä»Ž Keil/IAR
â·¥ç<è¼ççš»â° GCCi¼CEœèè¼ðæ³”æ,,ði¼š - Keil â°ð-âš”æ-†â»¶è€šä,ä,ðèf¼ç>æŽYèç«
GCC ä¼ç””â€, - GCC éœèè¼ð GNU æ±†ç¼¼-æ¼â¼¼â°ð-âš”æ-†â»¶â€, - ä,æ-
â°ð-è†ðè¼ç-†â°ð-â°ðè¼â°CE GD32 â°æ-†â»¶â€â,æ-â†¼æ°â°ðâ°ä¼æCEðä,€è†â€, ##
13. C æ †â††â°“æ”æ,,ðâ°è¼¹ Homebrew çš,, `arm-none-eabi-gcc` â°ð-èf¼ä,ðé™,,â¼â®ææ·

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newlib a0x-a0a0i1/4Ea0 a0c1/4-e0-a0-a0-e0f1/2a0c0Z0i1/4s00 text fatal error: stdint.h: No such
file or directory00 a0e-e0i1c0@R0c0s0,a0c0,c00+a0-a01/40a0-a0e00 firmware/include/
a0,a0e00a0/40a0e0a000e0f0,a0e0a00-a0-a0-a0i1/4s000 text firmware/include/stdint.h firmware/include/
stdbool.h00 a00Ea0-a0e004a0Z0Ya0-a0a0i1/2c000i1/4s000 text -nostdlib00 a0a0a0e004a0Z0 GCC a0+...
e0f0a00i1/4s000 text -lgcc00 e0i1c0-a00-a0Ya0-a0e00 CMSIS a0Ea01/20a000a00a0a0a00
E0e0e0c0s0,a0Y0c0i0c0a0a0z0i1/4Ea0,a0a0/40e0-a00Ea00 C a0a0a0a0a0a0e0,a0,a0,a0e0a00Z00e0e0e0e00
printf a0e00 malloc a0e00-a0-a0-a0i1c0 IO c00a00Ea00 C a00e0f1/2a0s0i1/4Ea00e0R0a00i1c000a0i1 newlib
c0s0, Arm GNU Toolchaini1/4Ea0-e0e0...a00c0-e0e0...0c01/20 newlib/newlib-nanoa0e0, ## 14.
a0z0,a00a00a0a0a0i1 e0i1a0...Ya0i1c0@R0a01c0@R0a01/2i1/4s000 sh cd ...00 e0...0c01/20a00Ya00i1/4s000 sh cmake -S
firmware -B firmware/build -G Ninja -DCMAKE_TOOLCHAIN_FILE=.../firmware/cmake/arm-
none-eabi-gcc.cmake -DCMAKE_BUILD_TYPE=Release00 c01/4-e00-a0Ya00i1/4s000 sh cmake --
build firmware/build00 a0-a0a0S0Ya0-a0Z0a0/4s0c0Y0a0-a0i1/4s000 text firmware/build/
gd32f303rct6_firmware.elf firmware/build/gd32f303rct6_firmware.bin firmware/build/
gd32f303rct6_firmware.hex firmware/build/gd32f303rct6_firmware.map00
a0e-e0i1c0@R0a01/20a000a0z0,a000c00a0a0z0e0i1/4s000 text FLASH: 25484 B / 256 KiB RAM: 4240 B / 48
KiB00 ## 15. c0f0a01/20a00a0a0a0i1 a0...0c0i1c0e0R0 ST-Linki1/4s000 sh st-info --probe00 c0f0a01/2000.bin0 a0-
a0-a0a0i1/4s000 sh st-flash --reset write firmware/build/gd32f303rct6_firmware.bin 0x080000000000
c0f0a01/20a0-a0a0S0Ya0-a0a0/4s0c0a00i1/4s000 text Flash written and verified! jolly good!00
a0i1,a0z0a0e000c00i1/4s000 text NRST is not connected --> using software reset via AIRCR00 e0-
a0Z0a0i1c0a0e000e0i1c0Z0Ya0i1-a0a0a00a01/200c0i1/4Ea00Ya0...0a00i1c000e01/20a0a0a00a01/20a0e0,a00e0i1c0a0a0TMa0...
Ya0Ea0 i0e0Ea0-a0a0S0Y0i1/4E0e0i1c0TMa0,a0a0-e0TM0e0-a0e0, ## 16. e00a00z0a0 i0e0E
a0,a0a0+0c0i1c0e0R0a00a00a0a0i1c0i1c0@R0a00Z0a0TMa0...Ya0e0c0i1c0i1/4Ea00-a0Ya00Z0e0S0c000 Flash e00a00z0i1/4s000 sh st-
flash read firmware/build/gd32f303rct6_firmware_readback.bin 0x08000000 300400
a0-a0a00a0e0-a0e0-a0-a0-a0-a0a0a0i1c0E00a00a0z0a0-a0-a0a0i1/4s000 sh cmp firmware/build/
gd32f303rct6_firmware.bin firmware/build/gd32f303rct6_firmware_readback.bin00 a0i1,a0z0a0
cmp0 a0i1c0a0e000e0e0/40a0a0i1/4E00-a0Z0a0,a0a0-a0-a0a0a0i1c0Ea0...0a0,e0e0a0e0, a0i1Ya0a0-a0Ya0e0R0i1c0R0-
SHA256i1/4s000 sh shasum -a 256 firmware/build/gd32f303rct6_firmware.bin firmware/build/
gd32f303rct6_firmware_readback.bin00 a0e-e0i1c0@R0a01/20a000a0a0 i0e0E0c00a0a0z0e0i1/4s000 text
cbee46a74aaf4cbf79b1123ec4cea892f6a5cdc9b86f91249bc92ca1f08114b0 firmware/build/
gd32f303rct6_firmware.bin
cbee46a74aaf4cbf79b1123ec4cea892f6a5cdc9b86f91249bc92ca1f08114b0 firmware/build/
gd32f303rct6_firmware_readback.bin00 ## 17. e0i1c0e0i1c0E0a0E0e0-
a01/20a0000c00a0a0/40a00a0a0a0a0i1c0E0a00Z0i1/4s0 - RGB LED a01/2c000 PC0a0e00PC1a0e00PC2a0e0, -
c0c0a0e00c00a0e0e0e0-0 LED a0-a00250 ms0 e0i1c00c000e01/20a0e0, - USART0 a01/2c000 PA9
TXa0e00PA10 RXa0e0, - a0a0a0f0a0,a00a000115200 8N1a0e0, -
a0a0a0f0e0/40a0a0a0a0-a0S0a0i1c0a0-a0Ea00a0e0Y0a0e0S0 ticka0e0, a0a0a0f0e0Z0a0Z0Ya0-a01/40i1/4s0 | USB-TTL |
a01/40a0a00a0e0i1c0 | --- | --- | RX | PA9 / USART0_TX | TX | PA10 / USART0_RX | | GND | GND |
a0a0a0f0a0Ya0...0a0-a0Ya0a01/2c000i1/4s0 - screen - minicom - picocom - macOS
a0a0/4a01/2c0a0a0a0f0a0Ya0...0c00a0/4i1/4s000 sh screen /dev/tty.usbserial-xxxx 11520000 ## 18. a0,e0s0e0-
R0e0c00### 18.1 a0e00/4a0,a0a00 arm-none-eabi-gcc0 a0f0e0a0Y0a0-a0a0i1c000e0f0...i1/4s000 sh brew list
--versions arm-none-eabi-gcc00 a0f0e0a0Y0a0a01/2a000e0-a0/4,i1/4s000 sh command -v arm-none-eabi-
gcc00 a0i1,a0z0a0e0a0i1c0a0e000e0e0/40a0a0i1/4Ea0f0e0a0Y0 Homebrew a0-a0a0a0S0a0...0Ya0 PATHa0e0, ### 18.2
CMake a0e00/4a0,a0a00a0Ya0...0e004a0-a0-a0a0a0i1c0E0a00e0R0a0a01/2c000c00a0a01e0-a0/4,i1/4s000 sh
-DCMAKE_TOOLCHAIN_FILE=.../firmware/cmake/arm-none-eabi-gcc.cmake00 ### 18.3
a0e00/4a0,a0a00 stdint.h0 a0Z0Y0a0 e0f0s0,a0-a01/20a000 Arm GCC a0i1c0a0e000a00Ea00 newlib a00-a0-
a0-a0a0i1c0a0e0, a0e-e0i1c0@R0a01/2c00a0e00a0a0/4i1/4s000 text firmware/include/stdint.h firmware/include/
stdbool.h00 a0i1,a0z0a0e0-a00a00Ya00i1/4E0e0e0e0e0i1c0a0i1c0c0TM0c0a0a0/4i1/4a0c0,c00i1/4Ea00-e0e0...a0R000e0f0...a0i1,
newlib c0s0,a0R0Ea00 Arm GNU Toolchaina0e0, ### 18.4 ST-Link a0e00/4a0,a0a00c00R0a0a0e0S0 c000a0 a0f0
Ea0Y0i1/4s0 - a01/40a0a00a0e0i1c0a0-a0a0a0/40c00a0f0, - SWDIO/SWCLK a0-a0a0a0Z0Ya01a0e0, - GND
a0-a0a0a0...0a0e0a0e0, - a0-a0a0i1c0e0e0e0e0e0i1c0Z0a0Z0Ya0 NRSTa0e0, - a00a0a0a0a0-a0a0a0...0e0-a0+0 SWDa0e0, ###
18.5 c0f0a01/20a00Z0a0-a0a0+0a0i1c0a0i1c0Z0a0Z0Ya0 a00-e0f1/2a0Z0Y0a0 i1/4s0 - c00a00a0a0a0+0 SWD/JTAG
a01/40e0,s0e0...0c01/20a0e0, - c00a000e0i1c0a0...Ya01/2Z0a0S0Y0e0-a0a0i1/40a0e0, - a0-a0e0Y0e0...
0c01/20e0TM0e0-a01/40e0e0S0c000a01/4,a0,a0e0, a00e0R0i1/4s0 - a00a0a0a0a0i1c0c0TM0 SWDa0e0, - e0i1c0Z0a0Z0Ya0
NRSTa0e0, - a0E000a01/20a0a0a01/20a0e00s0e0i1c0E0 st-info --probe0 a0-e0a0a0e0-c0f0a01/20a0e0, ## 19.
```

æŽ`è♦♦â·¥ä½œæµ ♦ æ—¥â,,â¼€â♦‘æŽ`è♦♦æŒ%ä»¥ä, <æµ♦ç`i¼š ``sh cd ... cmake --
build firmware/build st-flash --reset write firmware/build/gd32f303rct6_firmware.bin
0x08000000 `` éœèè|♦â@Œæ•‘é‡♦é...♦æ—¶i¼š ``sh cmake -S firmware -B firmware/build
-G Ninja -DCMAKE_TOOLCHAIN_FILE=.../firmware/cmake/arm-none-eabi-gcc.cmake
-DCMAKE_BUILD_TYPE=Release cmake --build firmware/build ``
éœèè|♦çj@è@çfšâ½•â†...â@¹æ—¶i¼š ``sh st-flash read firmware/build/
gd32f303rct6_firmware_readback.bin 0x08000000 3004 cmp firmware/build/
gd32f303rct6_firmware.bin firmware/build/gd32f303rct6_firmware_readback.bin `` ## 20.
æŒ»ç»“ Mac ä,Šé...♦ç½@ GD32 çŽ`âçfçš,,æ ,âçfæ~i¼š - ç”” `arm-none-eabi-gcc` æ»çä»£
Keil ç¼—è`‘â™™`ãŒ, - ç”” CMake/Ninja ç@jç♦†æŽ,,â»°ãŒ, -
ç””è‡ªâ@šä¹%œ“¾æŽ¥è,,šæœ—â@šä¹%œ Flash/RAMãŒ, - ç”” GNU â♦—âŠ`æ—
‡ä»¶â@Œæ^♦ä,Šç”µâ^♦âš<âŒ—ãŒ, - ç”” ST-Link â@Œæ^♦çfšâ½•â¹Œæ jéªŒãŒ, -
âŽÝâŽ, SDK â♦ªè`»â¼•ç””i¼Œä,♦ç>’æŽ¥äç@æ”¹ãŒ, â½“â%œ♦éj¹ç>@â²ç»♦â@Œæ^♦â·¥ä...
·é“¾â@%œèŒ...ãŒ♦â·¥ç`<æ♦â»°ãŒ♦âºä»¶æŽ,,â»°ãŒ♦ST-Link
çfšâ½•â¹Œæ`»â»žæ jéªŒi¼Œä♦—ä»¥ä½œä, °â♦Žç» GD32 èŒ,æœª¼€â♦‘çš,,âÿ°çjŒæ`jæ♦çãŒ,